IMAGE SUPER-RESOLUTION MODELS PERFORMANCE BENCHMARK REPORT

Empirical Benchmark: Bicubic Baseline vs. Deep Learning Models (Compact RTL & Software Architectures)

BENCHMARKED ARCHITECTURES & EVALUATED MODELS:

1. Bicubic (Baseline): Standard bicubic polynomial interpolation algorithm (non-parametric baseline).
2. **Compact SRCNN RTL: Hardware-compact architecture 1 -> 16 -> 8 -> 1 (1,649 parameters, INT8 Q7).**
3. **SRCNN Original: Original CNN baseline architecture 1 -> 64 -> 32 -> 1 (8,129 parameters, FP32).**
4. ESPCN: Efficient Sub-Pixel Convolutional Neural Network (PixelShuffle upscale in LR space).
5. FSRCNN: Fast Super-Resolution CNN with feature dimension shrinking and expanding layers.
6. VDSR: Very Deep Super-Resolution network with 20 convolutional layers and residual learning.
7. EDSR: Enhanced Deep Residual Networks (8 ResBlocks, 64 deep feature channels).
8. SRGAN: Super-Resolution Generative Adversarial Network optimized for perceptual quality.

REPORT OUTLINE (TABLE OF CONTENTS):

I. DATASET: SUB_NIH (NIH CHESTX-RAY14)

1. Magnification Factor: Scale 2x

Metric	Bicubic	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	40.0682 dB	39.3011 dB	39.6171 dB	37.2428 dB	31.9671 dB	**40.3761 dB**	40.3739 dB	37.5680 dB
SSIM ↑	0.9753	0.9668	0.9737	0.9610	0.9254	0.9777	**0.9783**	0.9636
MS-SSIM ↑	0.9976	0.9966	0.9967	0.9941	0.9724	0.9978	**0.9980**	0.9947
LPIPS ↓	0.0956	**0.0583**	0.1653	0.2581	0.4316	0.1140	0.0863	0.2118
NIQE ↓	9.9243	7.6964	6.9968	7.8967	10.3887	6.7623	**6.2820**	6.6020
EPI ↑	0.3274	0.3039	0.4451	0.1479	-0.0508	**0.4693**	0.4076	0.2412
MSE ↓	8.2065	9.0190	9.1015	15.1724	45.8111	**7.6720**	7.8904	14.7761
RMSE ↓	2.7030	2.8844	2.8460	3.6969	6.5953	**2.6124**	2.6330	3.6104
Latency (ms) ↓	15.33 ms	12.73 ms	44.61 ms	**4.80 ms**	8.11 ms	336.82 ms	111.59 ms	318.63 ms
Throughput (FPS) ↑	65.2 FPS	78.6 FPS	22.4 FPS	**208.5 FPS**	123.3 FPS	3.0 FPS	9.0 FPS	3.1 FPS

** Notes: (↑) Higher is better | (↓) Lower is better | Bold: Optimal metric value across benchmarked models*
Architectures: Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)

I. DATASET: SUB_NIH (NIH CHESTX-RAY14)

2. Magnification Factor: Scale 3x

Metric	Bicubic	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	37.5998 dB	38.4493 dB	39.2991 dB	36.2036 dB	18.9758 dB	39.2127 dB	**39.4904 dB**	37.1230 dB
SSIM ↑	0.9620	0.9538	0.9643	0.9423	0.6448	0.9647	**0.9658**	0.9489
MS-SSIM ↑	0.9936	0.9930	0.9941	0.9871	0.8146	0.9941	**0.9946**	0.9900
LPIPS ↓	0.2076	**0.1214**	0.2478	0.3216	1.0178	0.2363	0.2079	0.2945
NIQE ↓	9.9727	7.8957	7.7859	8.7711	27.3127	7.8913	**7.2587**	8.2229
EPI ↑	0.1663	0.1064	**0.2773**	0.0457	0.0043	0.2546	0.2249	0.1132
MSE ↓	14.9982	11.0340	9.4633	18.7101	852.7464	9.7603	**9.0453**	15.3459
RMSE ↓	3.6182	3.1787	2.9142	4.1291	28.9475	2.9520	**2.8508**	3.7284
Latency (ms) ↓	13.67 ms	12.87 ms	45.47 ms	**2.70 ms**	4.22 ms	395.91 ms	64.26 ms	241.86 ms
Throughput (FPS) ↑	73.2 FPS	77.7 FPS	22.0 FPS	**370.5 FPS**	236.8 FPS	2.5 FPS	15.6 FPS	4.1 FPS

** Notes: (↑) Higher is better | (↓) Lower is better | Bold: Optimal metric value across benchmarked models*
Architectures: Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)

I. DATASET: SUB_NIH (NIH CHESTX-RAY14)

3. Magnification Factor: Scale 4x

Metric	Bicubic	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	36.1873 dB	35.8637 dB	35.5253 dB	21.4287 dB	15.5623 dB	36.3261 dB	**36.4724 dB**	35.1208 dB
SSIM ↑	0.9495	0.9413	0.9501	0.6679	0.6252	0.9529	**0.9530**	0.9406
MS-SSIM ↑	0.9900	0.9893	0.9891	0.8780	0.8155	0.9906	**0.9911**	0.9859
LPIPS ↓	0.2865	**0.1773**	0.3408	0.6851	0.9235	0.3185	0.3000	0.3489
NIQE ↓	9.9859	8.4658	9.3520	16.9979	272.5003	8.8045	**8.2138**	8.9861
EPI ↑	0.0923	0.0309	**0.2054**	-0.0005	-0.0070	0.1309	0.1015	0.0835
MSE ↓	20.6699	21.5248	24.2238	484.6246	1874.2795	20.1630	**19.4813**	25.6018
RMSE ↓	4.2471	4.3670	4.5899	21.8243	42.8994	4.1883	**4.1170**	4.7600
Latency (ms) ↓	12.71 ms	12.73 ms	23.21 ms	**1.74 ms**	2.56 ms	368.23 ms	39.42 ms	195.89 ms
Throughput (FPS) ↑	78.7 FPS	78.6 FPS	43.1 FPS	**574.6 FPS**	390.1 FPS	2.7 FPS	25.4 FPS	5.1 FPS

** Notes: (↑) Higher is better | (↓) Lower is better | Bold: Optimal metric value across benchmarked models*
Architectures: Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)

II. DATASET: SUB_CHEST (CLINICAL CHEST X-RAY)

1. Magnification Factor: Scale 2x

Metric	Bicubic	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	**40.9900 dB**	39.2359 dB	39.5278 dB	37.4337 dB	32.3510 dB	39.9936 dB	40.3111 dB	38.1845 dB
SSIM ↑	**0.9597**	0.9308	0.9332	0.9078	0.8558	0.9432	0.9468	0.9119
MS-SSIM ↑	**0.9960**	0.9947	0.9921	0.9862	0.9542	0.9950	0.9957	0.9878
LPIPS ↓	0.1427	**0.1412**	0.2778	0.3876	0.5941	0.2027	0.1791	0.3399
NIQE ↓	9.9425	7.8182	6.9180	8.3889	9.7426	5.9437	**5.6151**	6.6622
EPI ↑	**0.3991**	0.2491	0.3511	0.0867	0.0002	0.3707	0.3718	0.2173
MSE ↓	**5.3800**	8.0486	7.6855	12.2547	38.8168	6.8945	6.4393	10.4249
RMSE ↓	**2.2965**	2.8108	2.7325	3.4636	6.1908	2.5895	2.4997	3.1849
Latency (ms) ↓	15.39 ms	492.30 ms	124.29 ms	**13.02 ms**	20.46 ms	861.29 ms	308.03 ms	799.78 ms
Throughput (FPS) ↑	65.0 FPS	2.0 FPS	8.0 FPS	**76.8 FPS**	48.9 FPS	1.2 FPS	3.2 FPS	1.3 FPS

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Architectures: Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)

II. DATASET: SUB_CHEST (CLINICAL CHEST X-RAY)

2. Magnification Factor: Scale 3x

Metric	Bicubic	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	**38.4502 dB**	37.5675 dB	38.1119 dB	35.8167 dB	19.4360 dB	38.0783 dB	38.3483 dB	36.8212 dB
SSIM ↑	**0.9338**	0.8993	0.9115	0.8779	0.6049	0.9132	0.9162	0.8836
MS-SSIM ↑	**0.9885**	0.9858	0.9859	0.9745	0.8140	0.9863	0.9874	0.9785
LPIPS ↓	0.2876	**0.2417**	0.4035	0.4873	1.0019	0.3675	0.3308	0.4806
NIQE ↓	9.9710	7.5179	7.8520	8.5688	21.9968	7.6123	**7.1297**	8.7333
EPI ↑	**0.2136**	0.0751	0.1927	0.0256	0.0004	0.1811	0.1767	0.1077
MSE ↓	**9.5621**	11.8406	10.5679	17.6864	748.9688	10.6449	10.0377	14.1808
RMSE ↓	**3.0694**	3.4070	3.2095	4.1663	27.2897	3.2216	3.1255	3.7202
Latency (ms) ↓	13.73 ms	444.88 ms	124.54 ms	**7.62 ms**	11.01 ms	934.05 ms	172.38 ms	558.56 ms
Throughput (FPS) ↑	72.8 FPS	2.2 FPS	8.0 FPS	**131.2 FPS**	90.8 FPS	1.1 FPS	5.8 FPS	1.8 FPS

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II. DATASET: SUB_CHEST (CLINICAL CHEST X-RAY)

3. Magnification Factor: Scale 4x

Metric	Bicubic	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	36.5994 dB	36.5373 dB	36.2227 dB	21.9458 dB	16.0441 dB	36.9682 dB	**37.1728 dB**	36.0685 dB
SSIM ↑	**0.9114**	0.8779	0.8898	0.6314	0.5881	0.8937	0.8945	0.8736
MS-SSIM ↑	**0.9823**	0.9795	0.9781	0.8779	0.8088	0.9802	0.9811	0.9724
LPIPS ↓	0.3955	**0.3138**	0.5324	0.7220	0.9233	0.4913	0.4775	0.5385
NIQE ↓	9.9821	**8.1220**	9.4849	14.7369	67.3225	8.7126	8.3824	9.2006
EPI ↑	**0.1297**	0.0310	0.1201	0.0009	-0.0003	0.1118	0.0864	0.0805
MSE ↓	14.5351	14.9630	16.2005	420.4435	1637.1846	13.6581	**13.0548**	16.8056
RMSE ↓	3.7918	3.8331	3.9820	20.4440	40.3374	3.6549	**3.5714**	4.0536
Latency (ms) ↓	12.66 ms	349.46 ms	60.57 ms	**6.61 ms**	8.51 ms	892.44 ms	135.01 ms	489.20 ms
Throughput (FPS) ↑	79.0 FPS	2.9 FPS	16.5 FPS	**151.2 FPS**	117.6 FPS	1.1 FPS	7.4 FPS	2.0 FPS

** Notes: (↑) Higher is better | (↓) Lower is better | Bold: Optimal metric value across benchmarked models*
Architectures: Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)